

Aircraft Pitch Control

MMAE 443

CAMERON HOOBER

KALEB MARTINY

LOGAN GARCIA





Introduction

Our team has designed an aircraft pitch control system and analyzed it using two different aircraft: A Boeing 747-100, and our crop duster from MMAE 414. All the stability derivatives were acquired from NASA CR-2144 "Aircraft Handling and Qualities Data" and from xflr5. The 747 system will be compared to the actual control system for the aircraft.



Model Development

SYSTEM DYNAMICS

Simplifying Assumptions

- Aircraft is steady, level, unaccelerated flight (SLUF)
 - Lift and weight; drag and thrust forces balance out
- Any deflection change (pitch) will not influence the speed of the aircraft
 - Constant velocity
- Aircraft is a rigid body
 - No change in shape/size
- No roll or yaw motion

Equations of Motion

Linearized short-period pitch-control model

$$\dot{\alpha} = \frac{Z_{\alpha}}{mV_0} \alpha + \left(\frac{Z_q}{mV_0} + 1 \right) q + \frac{Z_{\delta_e}}{mV_0} \delta_e$$

$$\dot{q} = \frac{M_{\alpha}}{I_{yy}} \alpha + \frac{M_q}{I_{yy}} q + \frac{M_{\delta_e}}{I_{yy}} \delta_e$$

$$\dot{\Theta} = q$$

Stability Derivatives

$$Z_{\alpha} = \frac{\partial Z}{\partial \alpha} = -\bar{q} S C_{L_{\alpha}} \quad M_{\alpha} = \frac{\partial M}{\partial \alpha} = \bar{q} S \bar{c} C_{m_{\alpha}}$$

$$Z_q = \frac{\partial Z}{\partial q} = -\bar{q} S \frac{\bar{c}}{2V_0} C_{L_q} \quad M_q = \frac{\partial M}{\partial q} = \bar{q} S \bar{c} \frac{\bar{c}}{2V_0} C_{m_q}$$

$$Z_{\delta_e} = \frac{\partial Z}{\partial \delta_e} = -\bar{q} S C_{L_{\delta_e}} \quad M_{\delta_e} = \frac{\partial M}{\partial \delta_e} = \bar{q} S \bar{c} C_{m_{\delta_e}}$$

Calculation Simplifications

$$A_1 = \frac{Z_{\alpha}}{mV_0} \quad A_2 = \left(\frac{Z_q}{mV_0} + 1 \right) \quad A_3 = \frac{Z_{\delta_e}}{mV_0} \quad B_1 = \frac{M_{\alpha}}{I_{yy}} \quad B_2 = \frac{M_q}{I_{yy}} \quad B_3 = \frac{M_{\delta_e}}{I_{yy}}$$

Constants For Different Aircraft

	$C_{L\alpha}$	$C_{m\alpha}$	C_{Lq}	C_{mq}	$C_{L\delta e}$	$C_{m\delta e}$	S (ft)	M (Slug)	$I_{yy} * 10^5$
747	5.67	-1.45	5.65	-21.4	.356	1.40	5500	22100	160
Crop Duster	5.03	-1.53	9.16	-13.8	-.62	1.90	758	542	3.052
	MAC (ft)	Vo (ft/s)	A1	A2	A3	B1	B2	B3	
	28.12	817	-.425	.99	-.267	-3.45	-.875	3.33	
	9.84	244	-1.76	.935	.2183	-2.28	-4.15	2.83	

General Transfer Function

$$\frac{\theta(s)}{\Delta(s)} = \frac{B_3s + (B_1A_3 - A_1B_3)}{s^3 - (B_2 + A_1)s^2 + (A_1B_2 - A_2B_1)s}$$

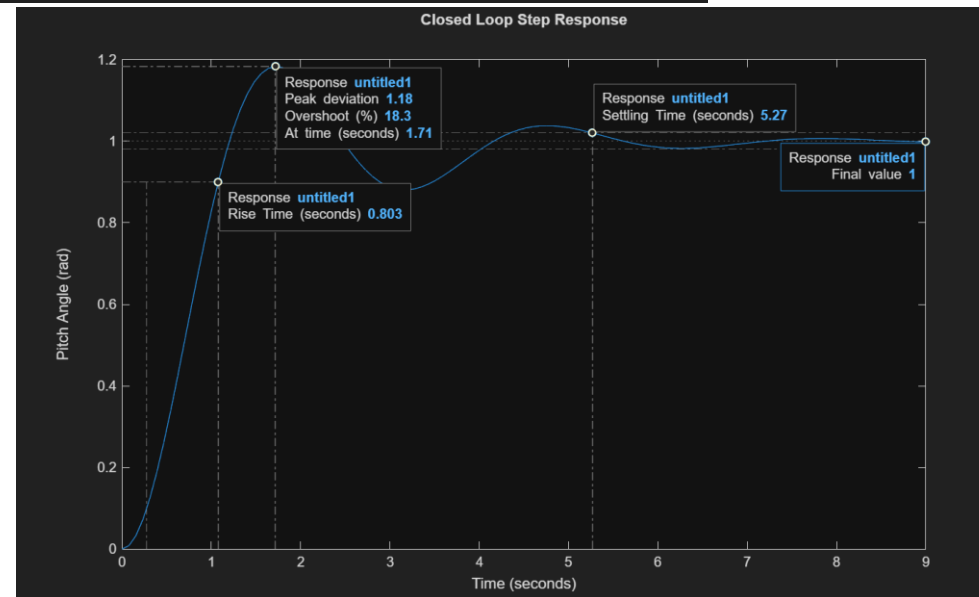
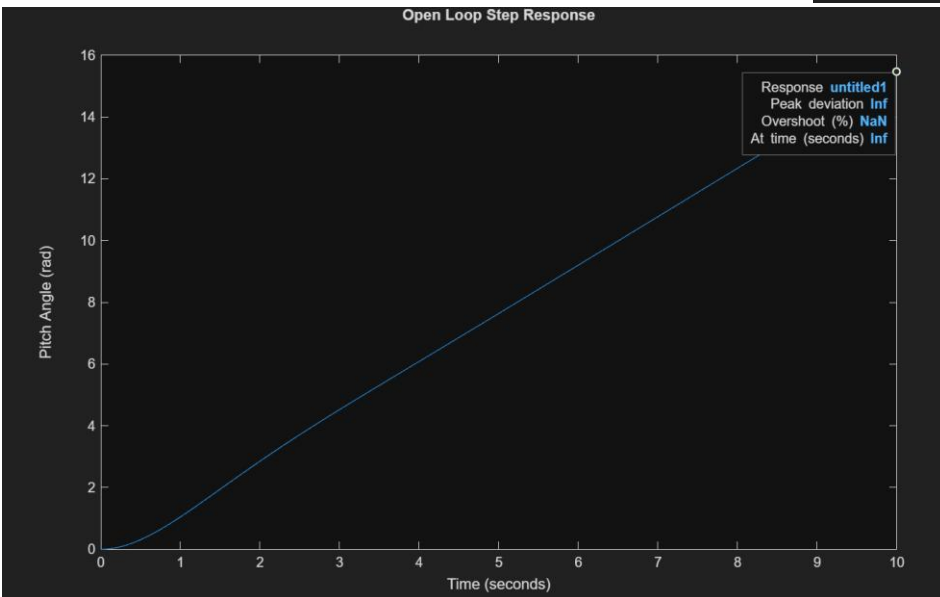


Analysis

STEP AND FREQUENCY
RESPONSES

Crop Duster - Step Response – 1 radian (57.3 degrees)

$$\frac{\theta(s)}{\Delta(s)} = \frac{2.8334s + 4.4887}{s^3 + 2.1736s^2 + 2.8650s}$$



Open Loop Poles:
 $0.0000 + 0.0000i$
 $-1.0868 + 1.2963i$
 $-1.0868 - 1.2963i$

Closed Loop Poles:
 $-0.5908 + 2.0430i$
 $-0.5908 - 2.0430i$
 $-0.9920 + 0.0000i$

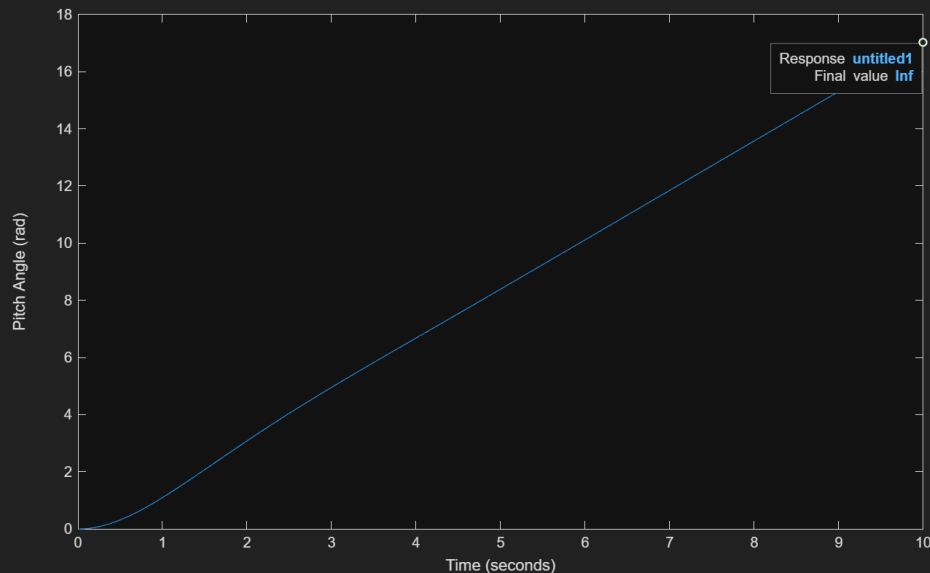
Closed Loop Zeros:
 -1.5834

Our base system is decently stabilized, but could be made better with PID controls
2 complex poles and 1 negative real pole (makes system stable)
Addition of feedback (closed loop) stabilizes the system compared to the open loop

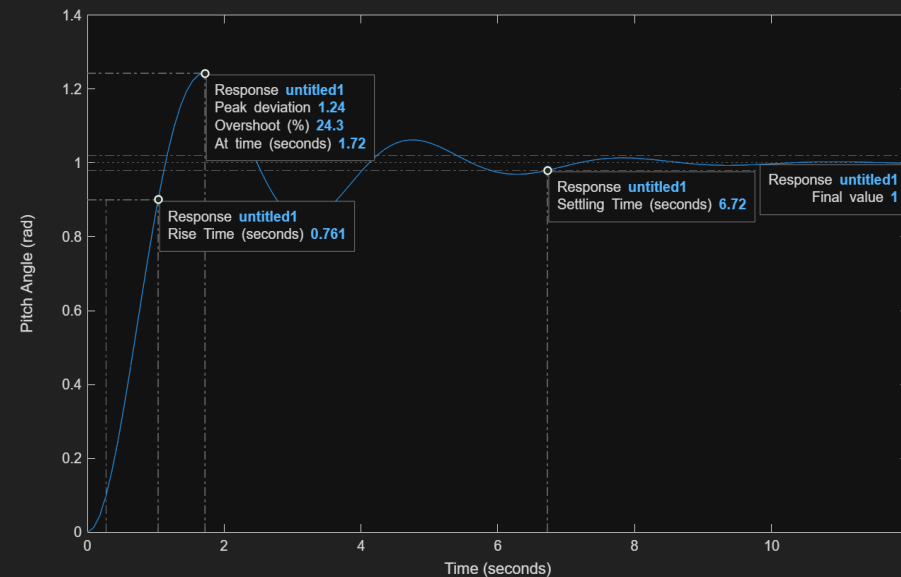
Crop Duster - Step Response Alteration #1 – Increase in pitching moment due to pitch rate B2 (-0.4146 ----> -0.263)

$$\frac{\theta(s)}{\Delta(s)} = \frac{B_3s + (B_1A_3 - A_1B_3)}{s^3 - (B_2 + A_1)s^2 + (A_1B_2 - A_2B_1)s}$$

Altered Open Loop Step Response



Altered Closed Loop Step Response



Altered Open Loop Poles:

0.0000 + 0.0000i
-1.0110 + 1.2541i
-1.0110 - 1.2541i

Altered Closed Loop Poles:

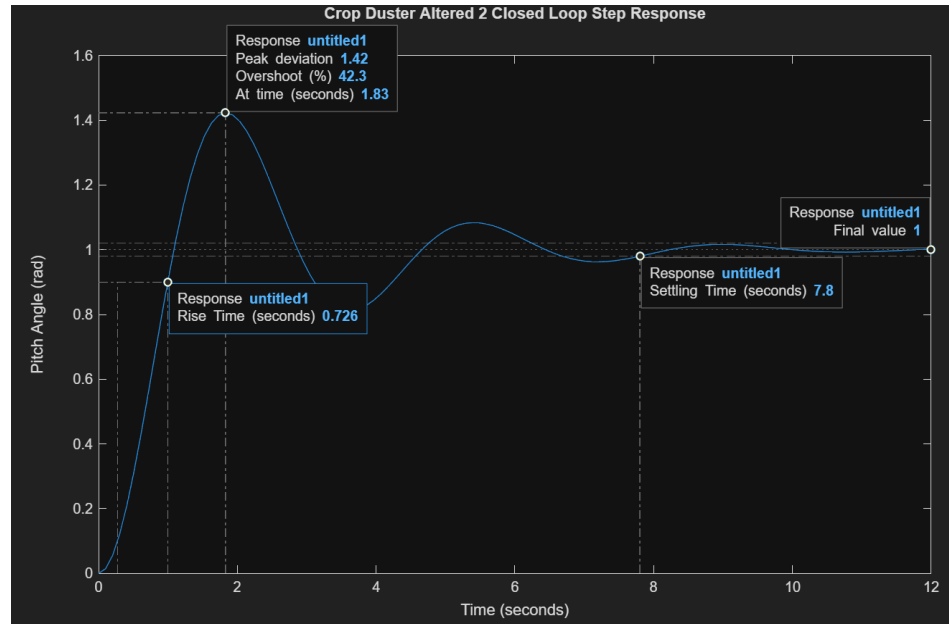
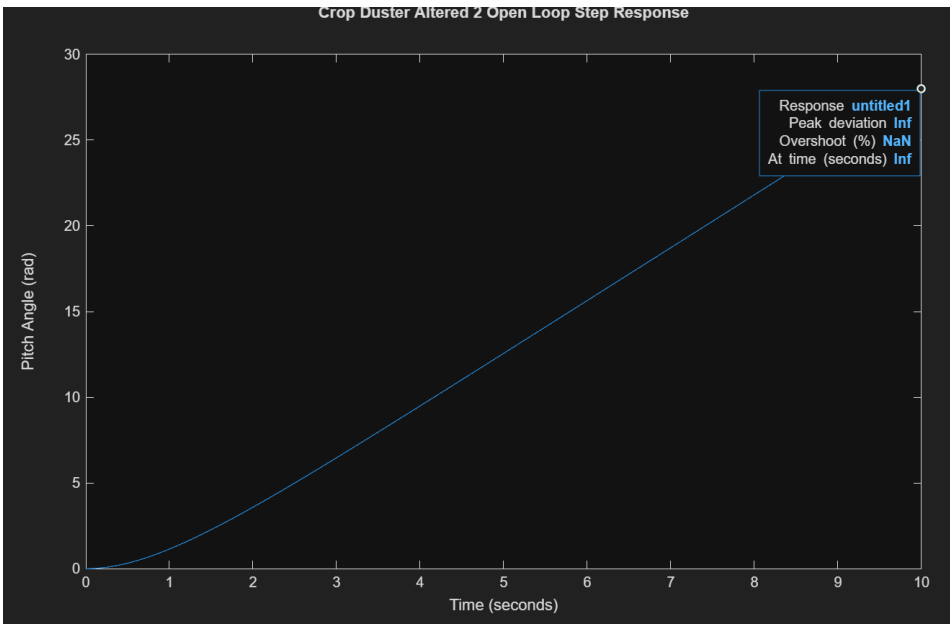
-0.5019 + 2.0382i
-0.5019 - 2.0382i
-1.0182 + 0.0000i

Altered Closed Loop Zeros:

-1.5834

Crop Duster - Step Response Alteration #2 - Decrease in angle of attack feeding back to alpha A1 (-1.7590 ----> -3.50)

$$\frac{\theta(s)}{\Delta(s)} = \frac{B_3s + (B_1A_3 - A_1B_3)}{s^3 - (B_2 + A_1)s^2 + (A_1B_2 - A_2B_1)s}$$



Altered 2 Open Loop Poles:

0
 -2.5795
 -1.1835

Altered 2 Closed Loop Poles:

-2.8567 + 0.0000i
 -0.4531 + 1.7584i
 -0.4531 - 1.7584i

Altered 2 Closed Loop Zeros:

-3.3244

Step Response Comparison

Increased the pitching moment due to the pitch rate B2: -0.4146 ----> -0.263

Decrease in angle of attack to alpha A1: -1.7590 ----> -3.50

Due to an alteration in B2 (increase in pitching moment due to pitch rate), which caused ONLY a change in the poles. This did not cause a major change in response but there were still differences.

Another alteration in A1 (decrease in AOA to alpha), manipulates both the poles and zeros, so there were more significant changes to the step response of the system making it more unstable.

$$G_{cl}(s) = \frac{G(s)}{1 + G(s)}$$

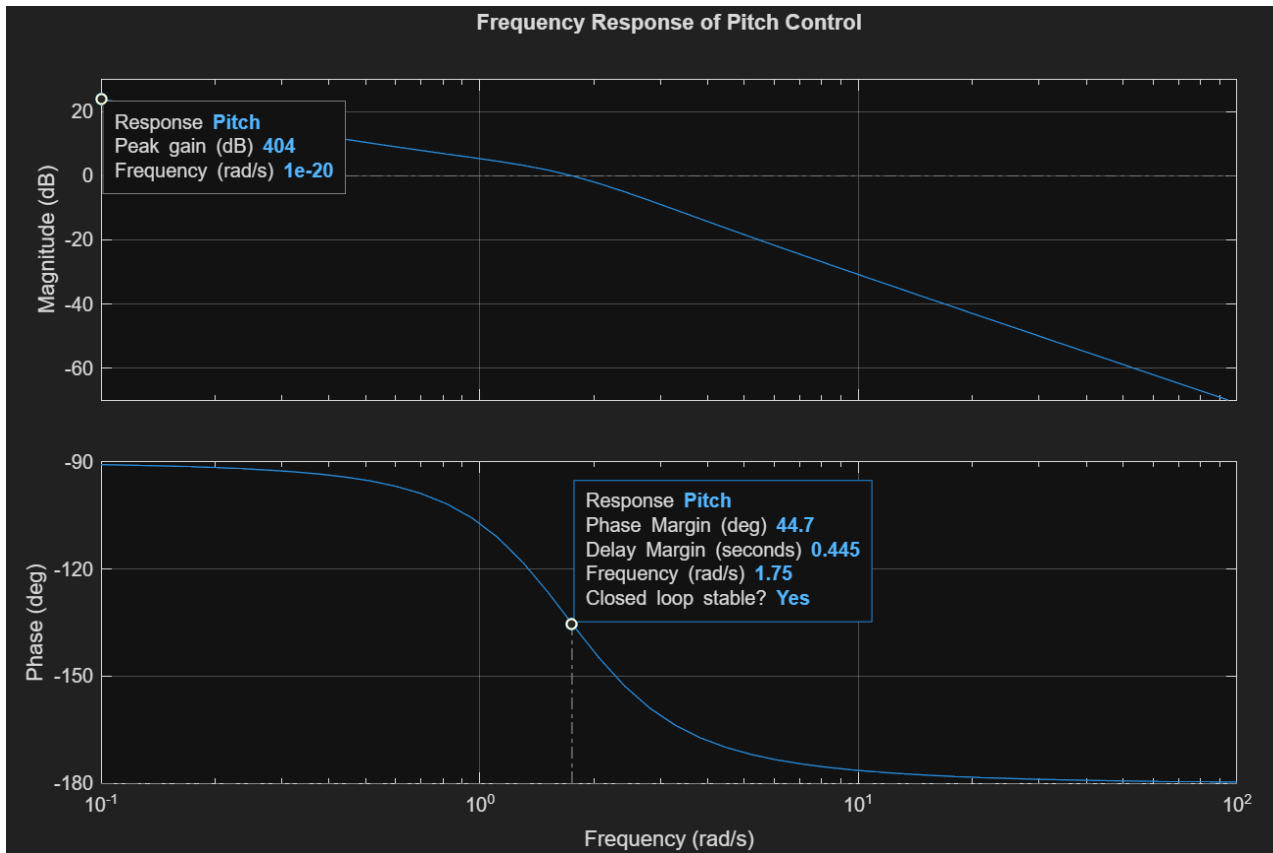
$$G_{cl}(s) = \frac{2.8334s + 4.4887}{s^3 + 2.1736s^2 + 5.6984s + 4.4887}$$

$$\frac{\theta(s)}{\Delta(s)} = \frac{2.8334s + 4.4887}{s^3 + 2.1736s^2 + 2.8650s}$$

Variable	Original	Alteration #1	Alteration #2
Rise Time (s)	0.803	0.761	0.726
Overshoot (%)	18.3	24.3	42.3
Settling Time (s)	5.27	6.72	7.8
Peak (rad)	1.18	1.24	1.42
CL Static Gain Gcl(0)	1	1	1
OL Static Gain	Inf (due to s=0)	Inf	Inf

Crop Duster - Frequency Response – Bode Plot

$$\frac{\theta(s)}{\Delta(s)} = \frac{2.8334s + 4.4887}{s^3 + 2.1736s^2 + 2.8650s}$$



Complex poles in the denominator cause negatively sloped plots:

Magnitude:

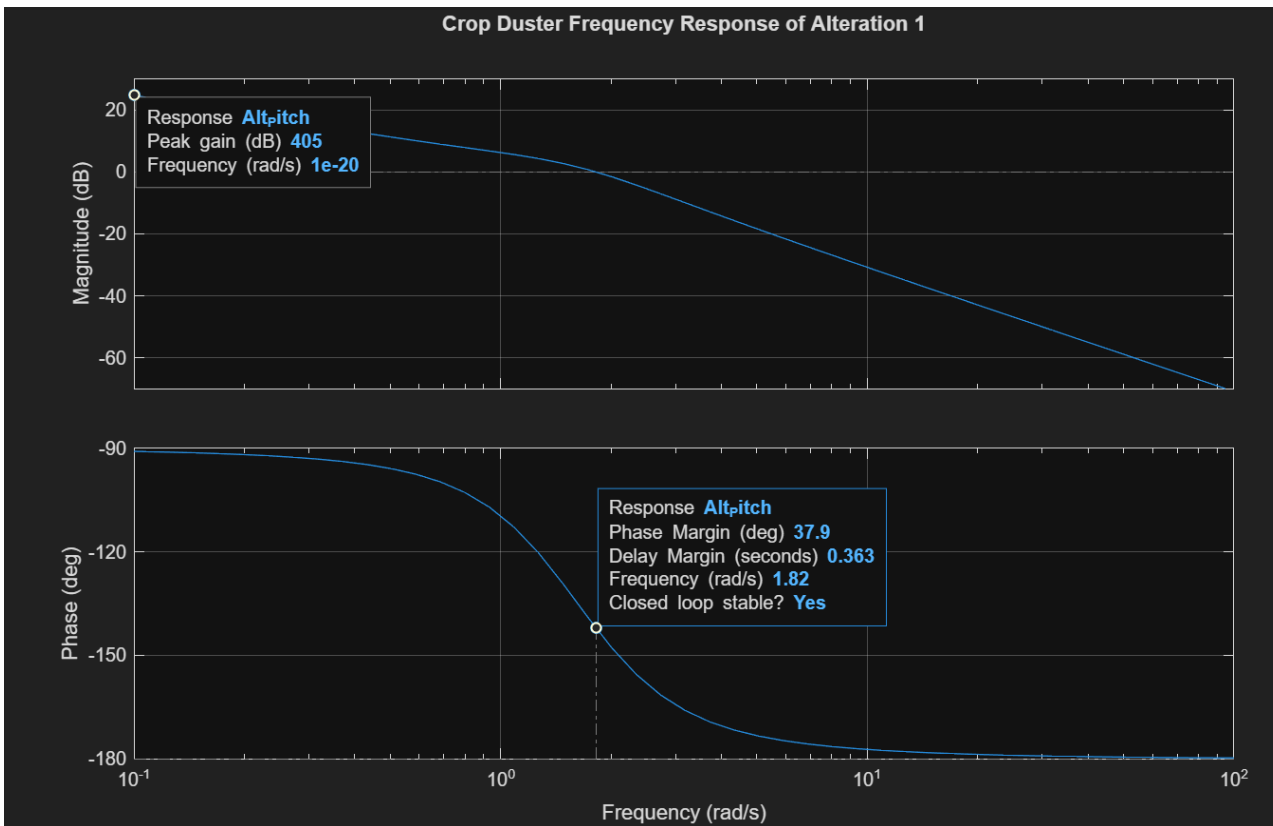
- At low freq the magnitude is about 20 dB
 - Due to constant in num and lowest term in denom
- At high freq slope is about -60 dB/dec
 - Due to 3rd order denom (around -40 dB/dec for 2nd order)
- **Break frequency** ~ natural frequency ~ 1.7 rad/s
- Damping ratio ~ 0.64 so there's a slight resonant peak since it's less than 0.707
- 1/s pole in denom has -20dB/dec from 0 to 1 rad/s

Phase:

- At low freq the pole at 0 starts phase at -90 deg
- Approaches -180 deg due to complex poles in denom

Crop Duster - Frequency Response – Bode Plot Alteration 1

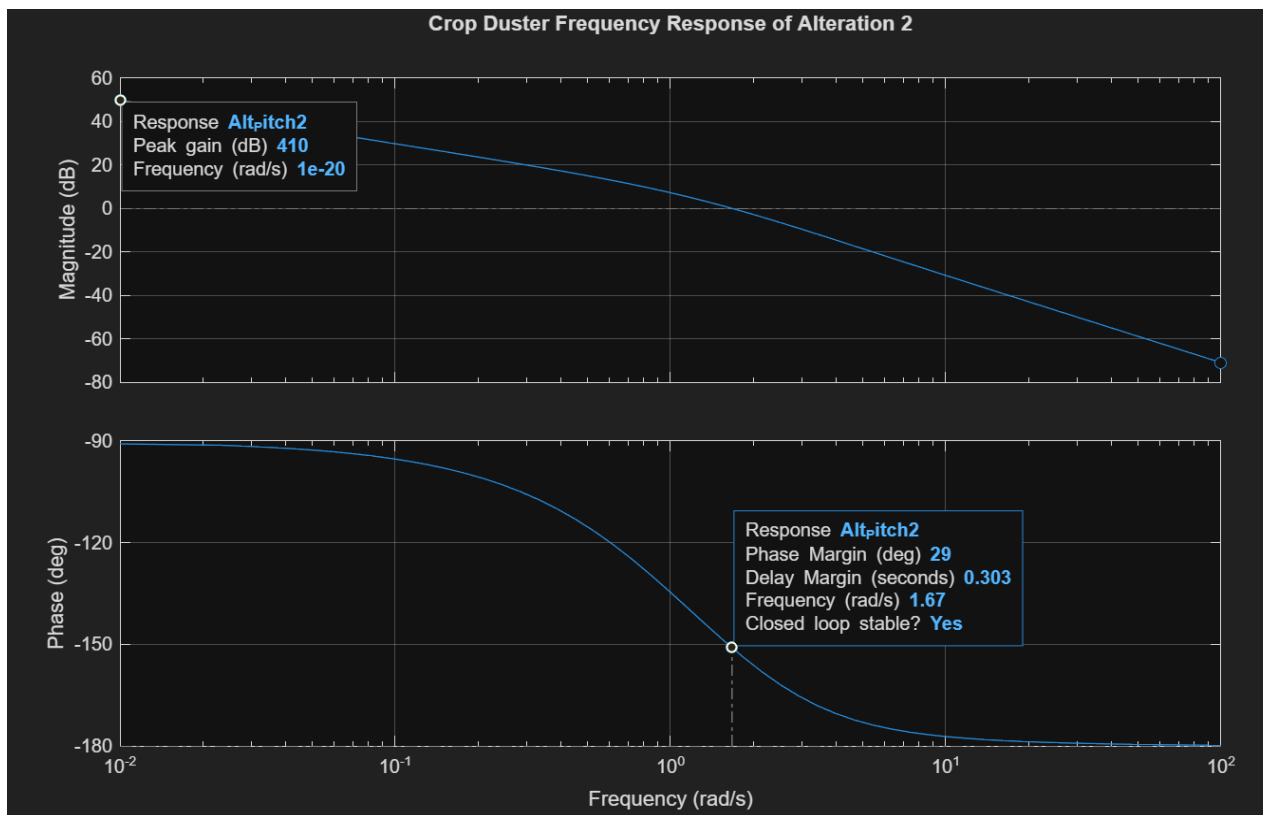
$$\frac{\theta(s)}{\Delta(s)} = \frac{B_3s + (B_1A_3 - A_1B_3)}{s^3 - (B_2 + A_1)s^2 + (A_1B_2 - A_2B_1)s}$$



- Not a significant difference
- The increase in the pitching moment due to the pitch rate affected the stability margin that shifted the **gain crossover frequency higher** and:
 - Phase margin (deg) ---> decreased
 - Delay margin (s) ---> decreased
 - Frequency at Margin (rad/s) ---> increased
 - Peak gain (dB) ---> increased

Crop Duster - Frequency Response – Bode Plot Alteration 2

$$\frac{\theta(s)}{\Delta(s)} = \frac{B_3s + (B_1A_3 - A_1B_3)}{s^3 - (B_2 + A_1)s^2 + (A_1B_2 - A_2B_1)s}$$



With a significant change in A1, there was a large difference in bode plots comparing to the original stability margin due to making the system less stable when poles AND zeros changed:

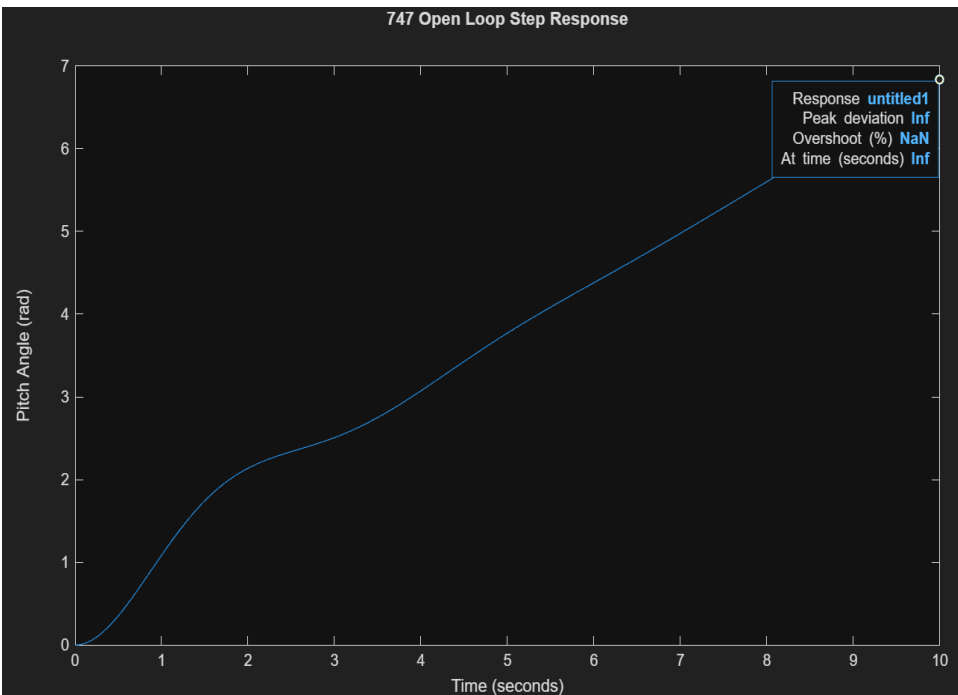
- Phase margin (deg) ---> decreased
- Delay margin (s) ---> decreased
- Frequency at Margin (rad/s) ---> decreased
- Peak gain (dB) ---> increased

Bode Plot Comparison of the Crop Duster

Variable	Original	Alteration #1	Alteration #2
Phase Margin (deg)	44.7	37.9	29
Delay Margin (s)	0.445	0.363	0.303
Frequency at margin (rad/s)	1.75	1.82	1.67
Peak Gain (dB)	404	405	410

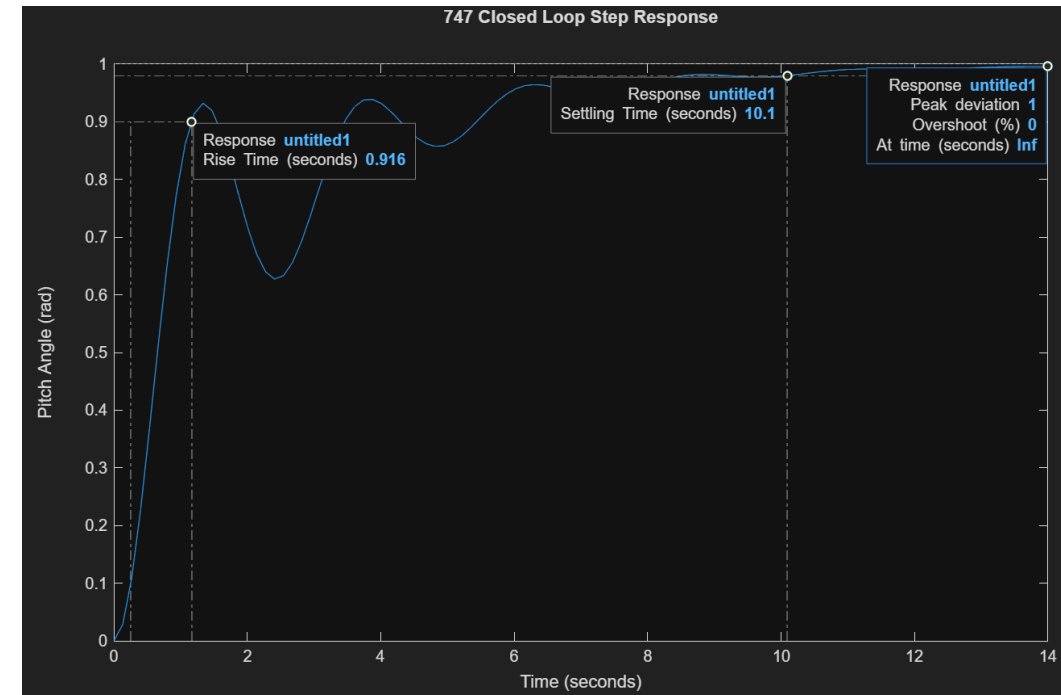
- All three phase plots do NOT crossover at -180 deg so the gain margin is undefined or infinite
- There is crossover at 0 dB for all three amplitude plots, so the phase margin is defined

Boeing 747 – 1 rad (57.3 deg) step response



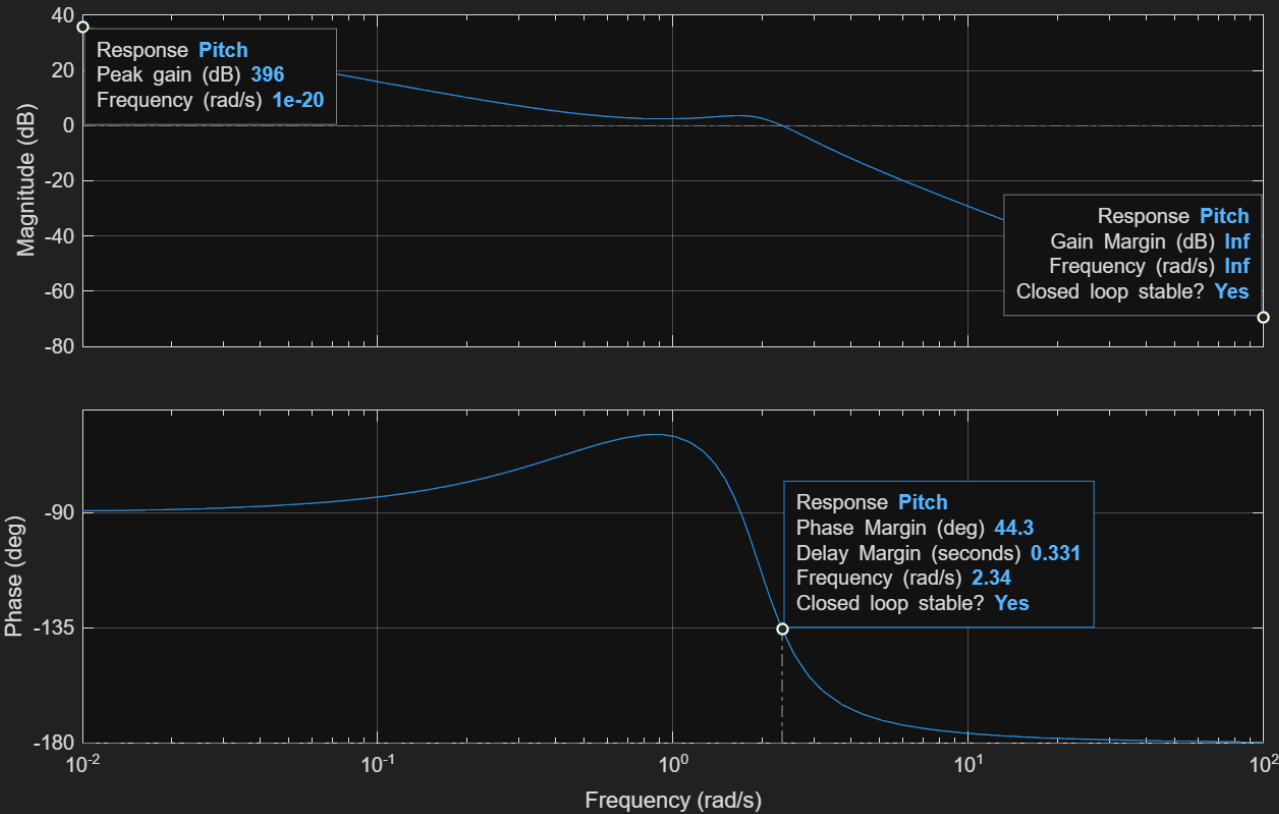
%OS = 0%
Ts = 10.1 sec
Rise Time = 0.916 sec

System is well
stabilized with no
overshoot



Boeing 747 Bode Plot

747 Frequency Response of Pitch Control



Peak Gain = 396 dB

Phase Margin = 44.3 deg

Delay Margin = 0.331 sec

Frequency at Margin = 2.34 rad/s

The Boeing 747 pitch control is a known stable system

- Crossover at 0 dB in amplitude plot, so the phase margin is defined
- Infinite gain margin
 - No amount of gain can cause closed loop system to become unstable

Crop Duster vs. Boeing 747

Bode Plots

Variable	Crop Duster	Boeing 747
Phase Margin (deg)	44.7	44.3
Delay Margin (s)	0.445	0.331
Frequency at margin (rad/s)	1.75	2.34
Peak Gain (dB)	404	396

Step Response Plots

Variable	Crop Duster	Boeing 747
% OS	18.3	0
Ts (sec)	5.27	10.1
Rise Time (sec)	0.803	0.916

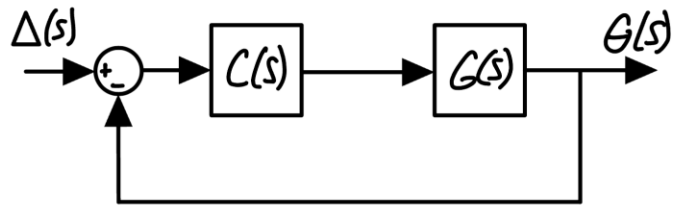
The Crop Duster may have a quicker response system, but it has a higher overshoot percentage where the Boeing does not, bringing more instability to the overall system



Crop Duster PID Control Design

SETTLING TIME,
OVERSHOOT, STEADY STATE
ERROR, ROOT LOCUS

PID Controller Model and Design Requirements



$$\frac{\left(k_p + k_d s + k_I \frac{1}{s}\right) \left(\frac{(2.8334s + 4.4865)}{s^3 - 2.1776s^2 + 2.86165}\right)}{1 + \left(k_p + k_d s + k_I \frac{1}{s}\right) \left(\frac{(2.8394s + 4.4865)}{s^3 - 2.1776s^2 + 2.86165}\right)}$$

Settling Time	Percent Overshoot	Steady State Error
< 15 seconds	< 5%	0

- Settling Time: Sub-15 seconds was chosen because the response of a pitching maneuver for an aircraft is not instant but is still expected to happen shortly after the input command is given. This time also allows for normal perturbations to be accounted for.
- Percent Overshoot: Less than 5% was chosen so that the pitch response of the aircraft is sufficiently damped and so that the oscillations following are not severe, allowing for good control of the aircraft.
- Steady State Error: 0 steady state error was chosen because when a pilot or autopilot inputs a command to climb at some angle of attack, it is essential that the aircraft does exactly what is commanded. This is for the safety of both the pilot/plane, as well as other planes that may be in the area.

PID Controller Coefficients

Expanded Closed Loop Characteristic Polynomial Using Actual Aircraft Coefficients and a PID Controller

$$s^4 + (2.8334 \times D - 2.1736) \times s^3 + (4.4865 \times D + 2.8334 \times P + 2.616) \times s^2 + (2.8334 \times I + 4.4865 \times P) \times s + 4.4865 \times I = 0$$

Gain were found using Routh-Hurwitz Stability Criteria and a MATLAB Script

Routh Array

s^4	1	B	D
s^3	A	C	0
s^2	$\frac{A \times B - C}{A}$	D	0
s^1	$\frac{C \times (A \times B - C) - A^2 \times D}{A \times B - C}$	0	0
s^0	D	0	0

First Column Must Be > 0

$$A = 2.8334 \times D - 2.1736$$

$$B = 4.4865 \times D + 2.8334 \times P + 2.616$$

$$C = 2.8334 \times I + 4.4865 \times P$$

$$D = 4.4865 \times I$$

Routh Inequalities and PID Coefficients

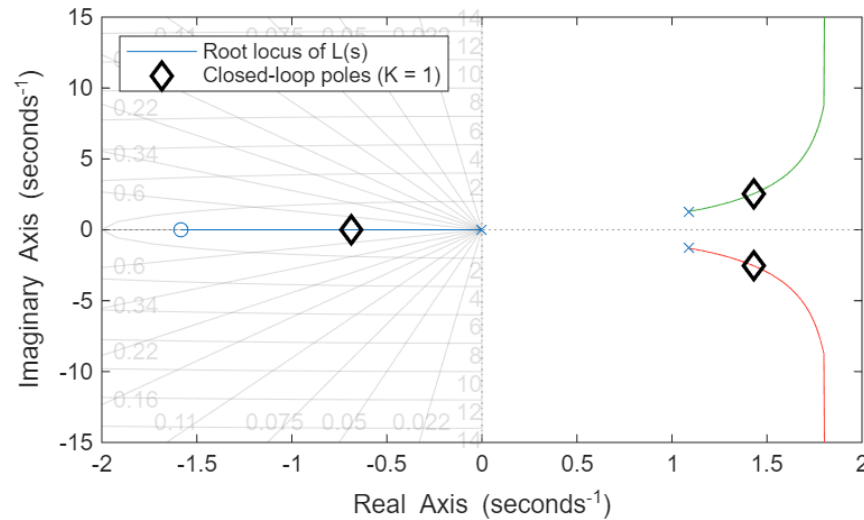
P	< 1.3113	1.3
I	> 0	0.4
D	> 0.767	1.2

Actual Design Outputs

Settling Time	< 15 Seconds	13.27 Seconds
Overshoot	$< 5\%$	2.56%
Error	0	0

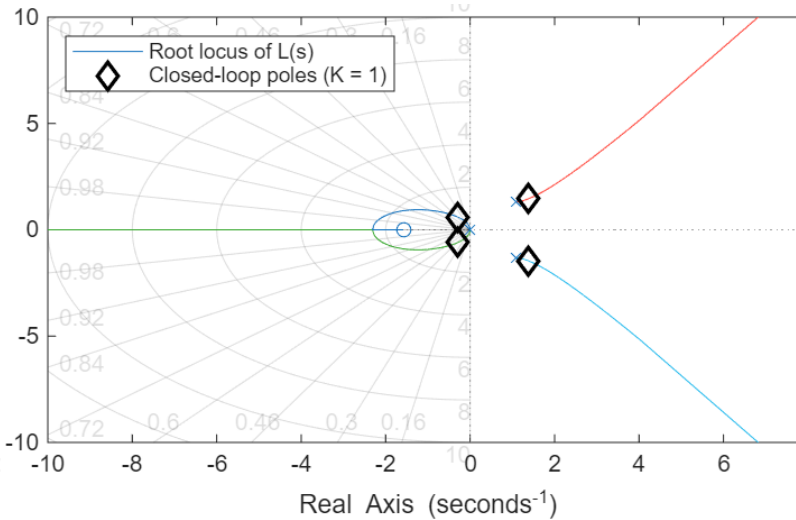
PID Gain Effects on Root Locus

P Only



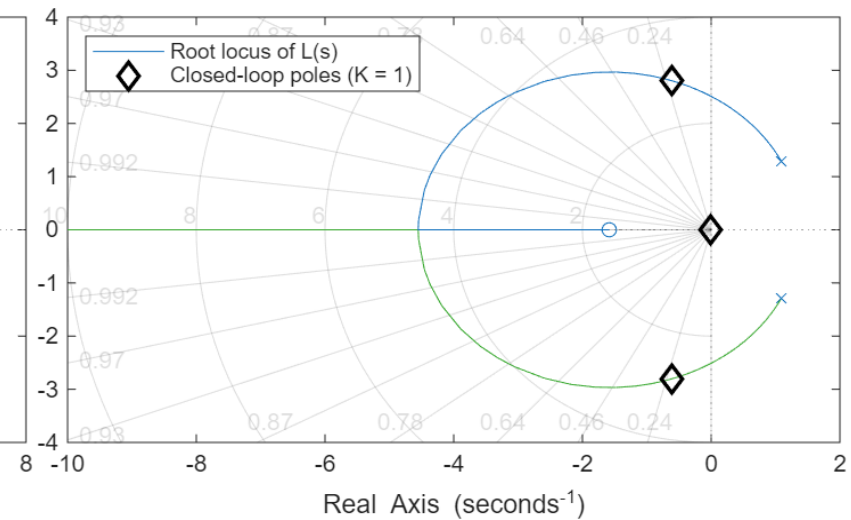
P shifts the locations of the closed loop poles along the locus paths

I Only



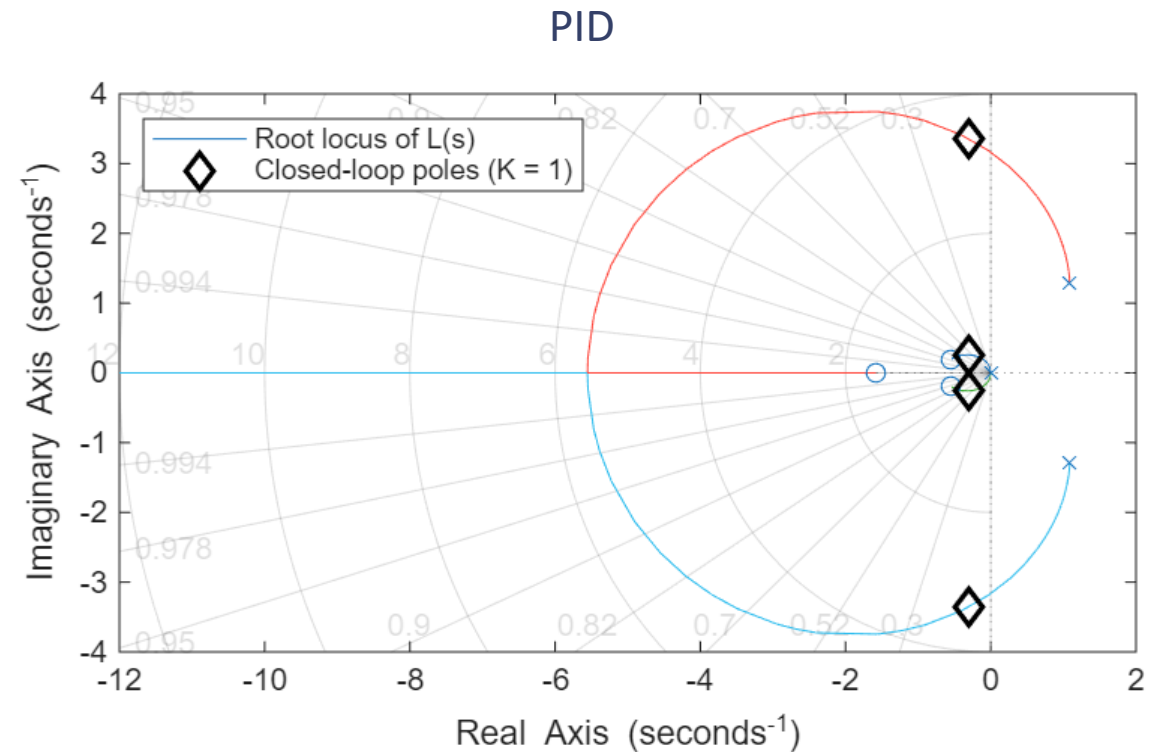
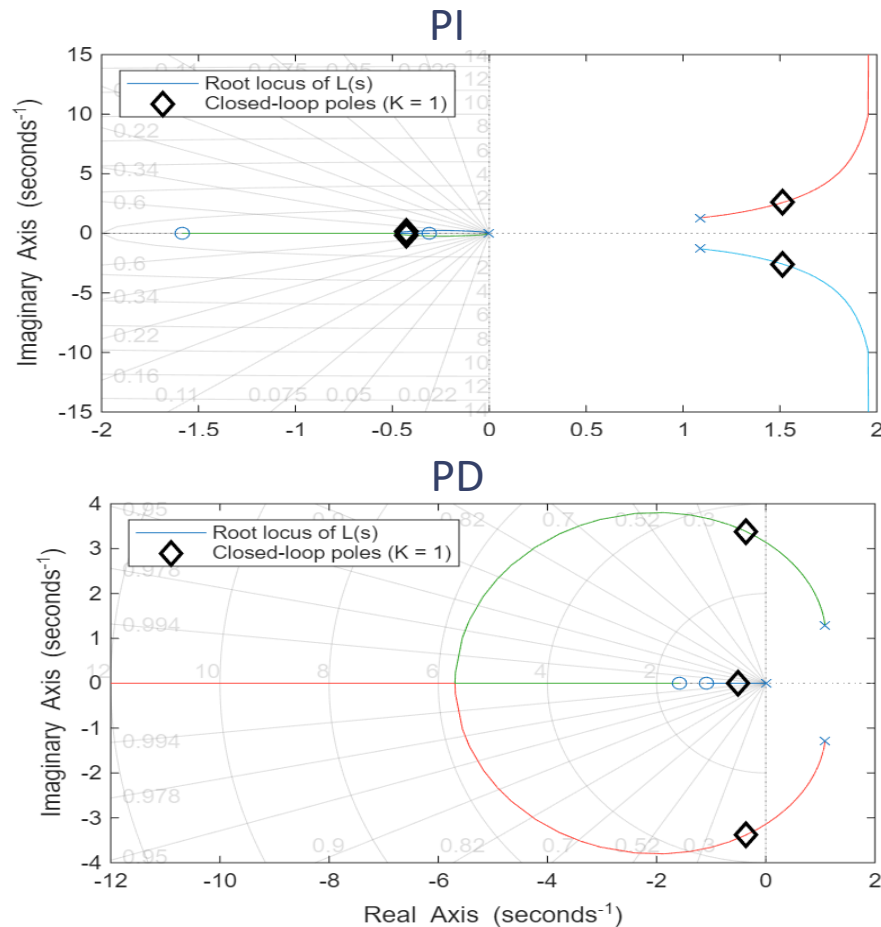
I adds a new pole at the origin because of $\frac{1}{s}$

D Only



Adds a new zero and pulls the poles leftward

PID Gain Effects on Root Locus



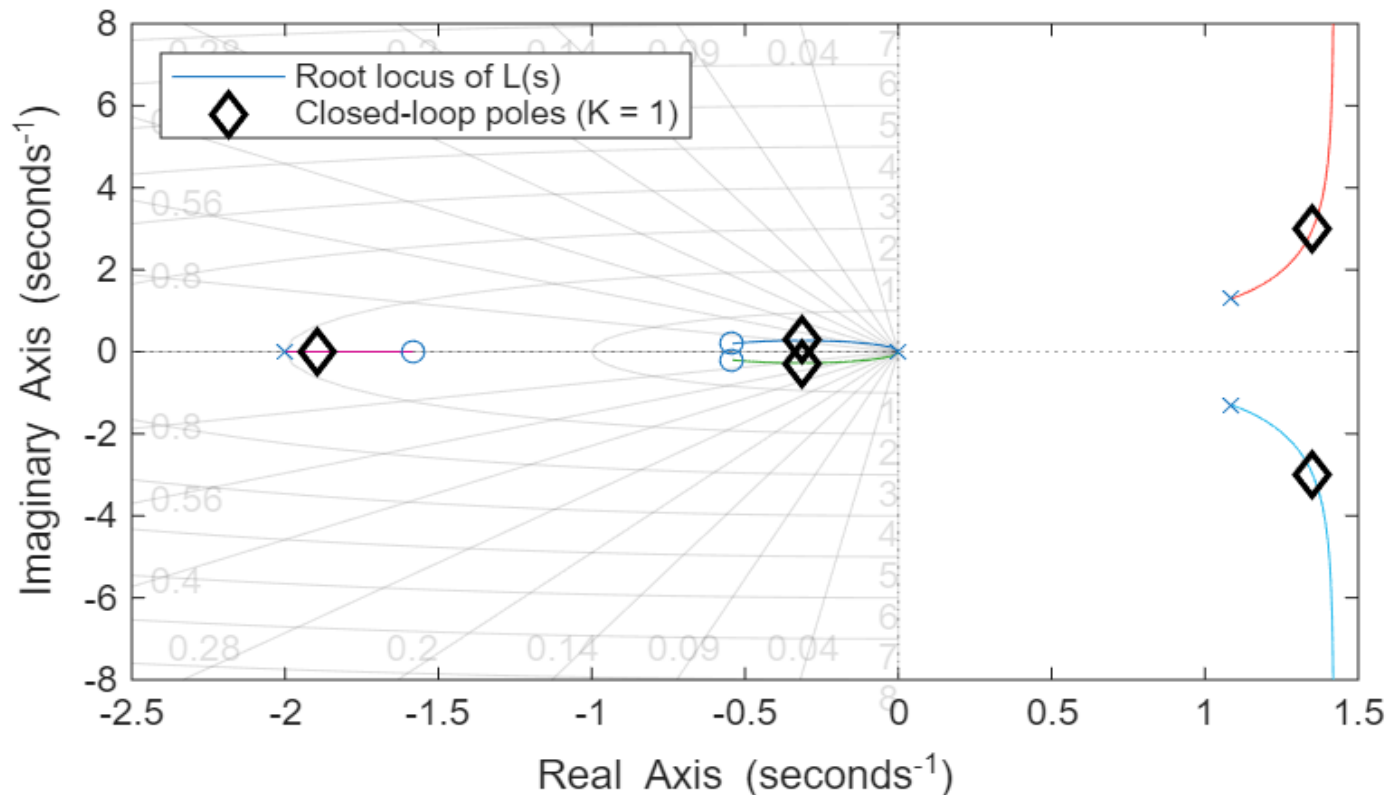
Closed Loop Poles

-0.3119
 $\pm 3.3613i$

-0.3013
 $\pm 0.2528i$

Addition of Sensor Dynamics

Root Locus with Sensor Dynamics



Design Output with Sensor Dynamics

Settling Time	< 15 Seconds	2.96 Seconds
Overshoot	< 5%	413.42%
Error	0	0

Closed Loop Poles

$$-0.3168 \pm 0.2746i$$

$$1.3513 \pm 2.9911i$$

$$-1.8954 \pm 0.0000i$$



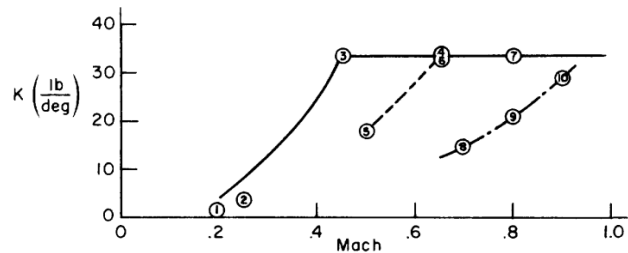
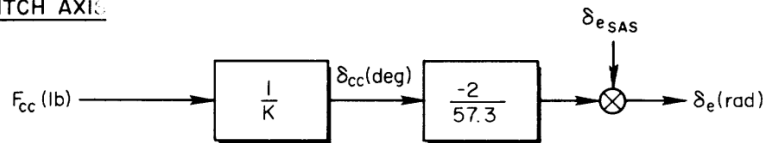
747-100 PID Control Design

SETTLING TIME,
OVERSHOOT, STEADY STATE
ERROR, ROOT LOCUS

747 PID Controller Model and Design Requirements

B-747

PITCH AXIS



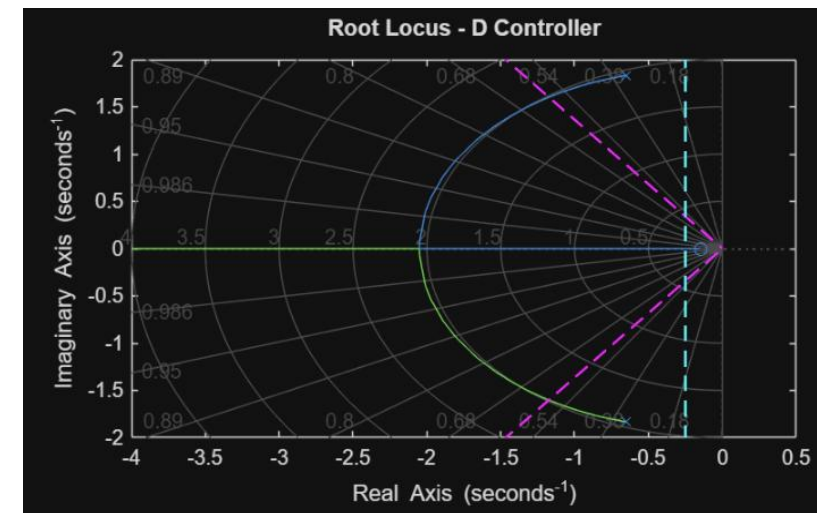
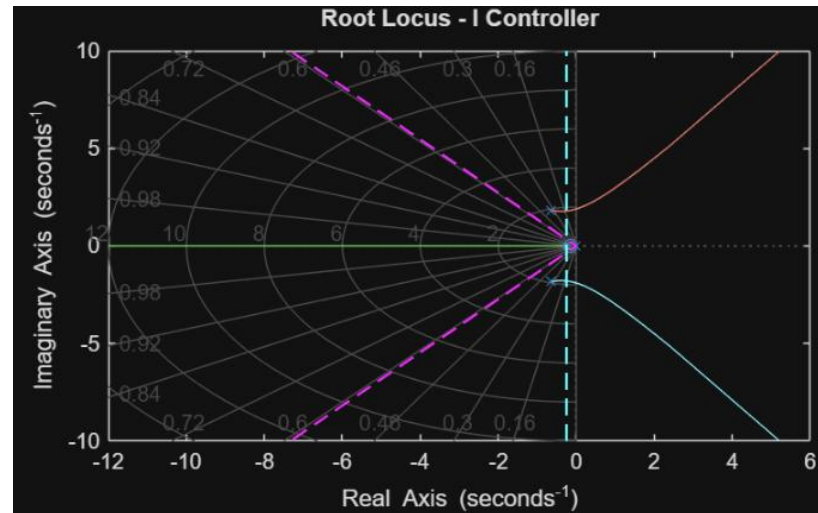
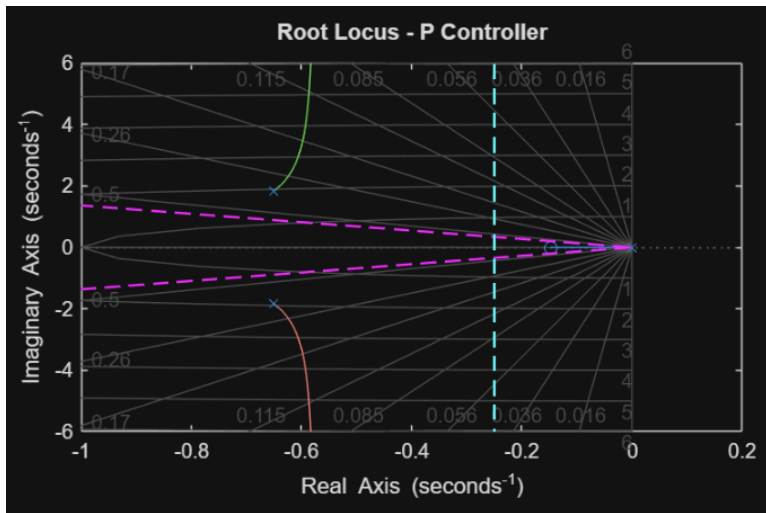
Settling Time	Percent Overshoot	Steady State Error
< 16 seconds	< 10%	0

These values are chosen as the 747 is a large civilian aircraft you want the plane to be slow, smooth, and well damped. Smaller aircraft like the crop duster want to be responsive and snappy as it needs to make tight turns and maneuvers.

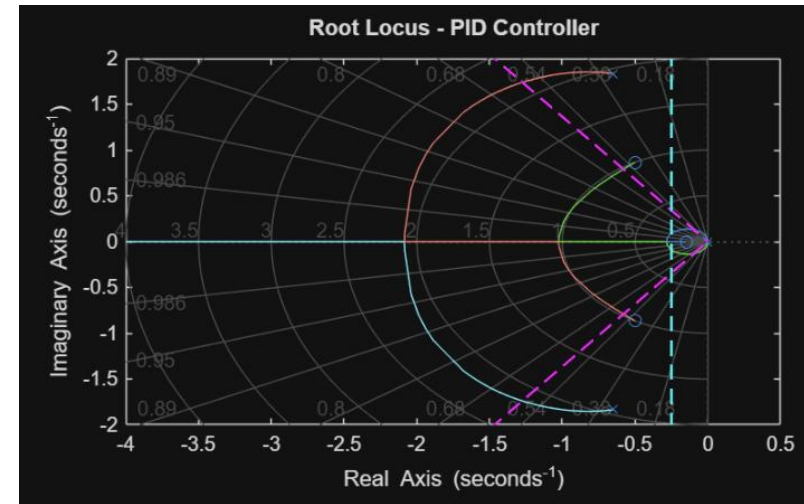
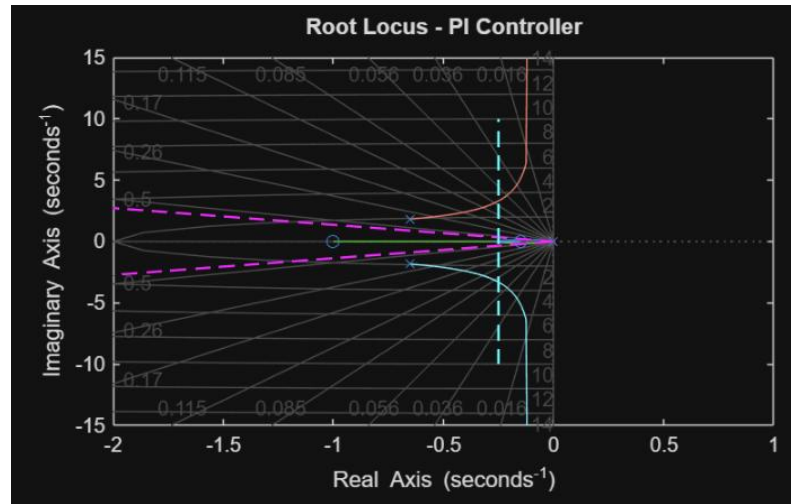
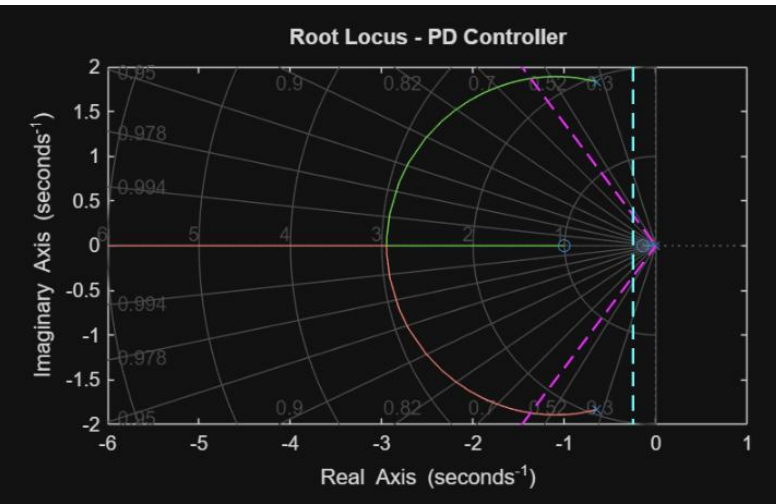
Assume elevator SAS is a closed-loop PID Controller and Mach = .8 $\rightarrow K = 32(\text{lb/deg})$

$$\frac{\left(\frac{-2}{57.3K}\right) \left(\frac{3.33s + 2.34}{s^3 + 1.30s^2 + 3.79s}\right)}{1 + \left(k_p + k_d s + k_I \frac{1}{s}\right) \left(\frac{3.33s + 2.34}{s^3 + 1.30s^2 + 3.79s}\right)}$$

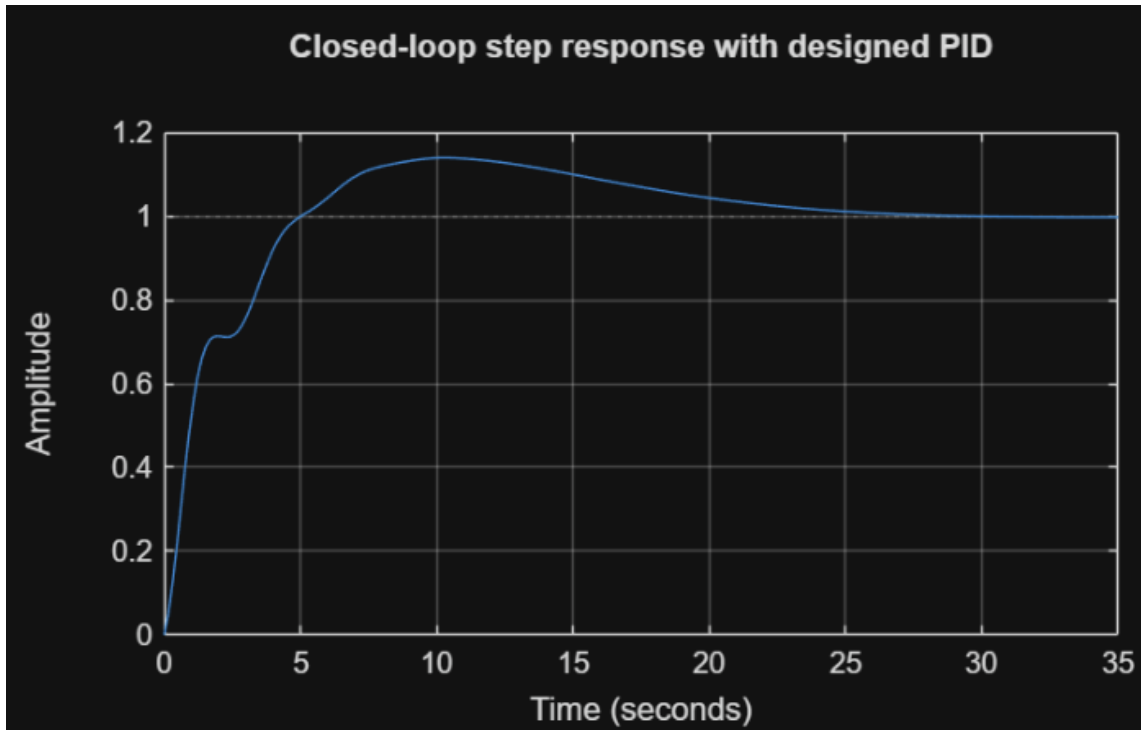
747 PID Controller Root Locus Plots



747 PID Controller Root Locus Plots



747 PID Controller



$$K_p + K_i * \frac{1}{s} + K_d * s$$

with $K_p = 437$, $K_i = 397$, $K_d = 64.5$

```
S = struct with fields:
    RiseTime: 3.5961
    TransientTime: 23.2568
    SettlingTime: 23.2568
    SettlingMin: 0.9164
    SettlingMax: 1.1398
    Overshoot: 13.9752
    Undershoot: 0
    Peak: 1.1398
    PeakTime: 10.2196
```



```
if success() == True:  
    celebrate()  
while success() == False:  
    try_again()  
    be_awesome()
```

MATLAB Scripts

Bode Plot Analysis, Frequency
Response, Step Response, PID
Control

Step Response and Bode Plot Scripts

Crop Duster (first 2 images) ; Boeing (3rd image)

% Agricultural Aircraft

A1 = -1.7590; A2 = 0.9354; A3 = 0.2182;
B1 = -2.2796; B2 = -0.4146; B3 = 2.8334;

s = tf('s');
Pitch = (((B3*s) + (B1*A3 - A1*B3)) / ((s^3) - ((B2+A1)*s^2) + ((A1*B2 - A2*B1)*s)));
STP = 1; % 1 rad (57.3 deg) step response ; input amount of step in radians

% Open Loop Step Response

t = [0:0.01:10]; %10 sec run time

figure;
step(STP*Pitch,t);
axis([0 10 0 0.8]);
ylabel('Pitch Angle (rad)');
title('Crop Duster Open Loop Step Response');
ol_poles = pole(Pitch);
disp('Open Loop Poles:'); disp(ol_poles)

% Closed Loop Step Response

cl = feedback(Pitch,1);
figure;
step(STP*cl);
ylabel('Pitch Angle (rad)');
title('Crop Duster Closed Loop Step Response');
cl_poles = pole(cl);
cl_zeros = zero(cl);
disp('Closed Loop Poles:'); disp(cl_poles);
disp('Closed Loop Zeros:'); disp(cl_zeros);

k = dcgain(cl);
k2 = dcgain(Pitch);
disp('OL Static Gain'); disp(k2);
disp('Static Gain'); disp(k);

figure;
bode(Pitch)
grid on
title('Crop Duster Frequency Response of Pitch Control')

% Changing a variable in denominator (B2 ----> B4)

A1 = -1.7590; A2 = 0.9354; A3 = 0.2182;
B1 = -2.2796; B4 = -0.263; B3 = 2.8334;

s = tf('s');
Alt_Pitch = (((B3*s) + (B1*A3 - A1*B3)) / ((s^3) - ((B4+A1)*s^2) + ((A1*B4 - A2*B1)*s)));
STP = 1; % 1 rad (57.3 deg) step response input amount of step in radians

% Open Loop Step Response

t = [0:0.01:10]; %10 sec run time

% Open Loop Step Response

t = [0:0.01:10]; %10 sec run time

figure;
step(STP*Alt_Pitch,t);
axis([0 10 0 0.8]);
ylabel('Pitch Angle (rad)');
title('Crop Duster Altered 1 Open Loop Step Response');
ol_poles = pole(Alt_Pitch);
disp('Altered 1 Open Loop Poles:'); disp(ol_poles)

% Closed Loop Step Response

cl = feedback(Alt_Pitch,1);
figure;
step(STP*cl);
ylabel('Pitch Angle (rad)');
title('Crop Duster Altered 1 Closed Loop Step Response');
cl_poles = pole(cl);
cl_zeros = zero(cl);
disp('Altered 1 Closed Loop Poles:'); disp(cl_poles);
disp('Altered 1 Closed Loop Zeros:'); disp(cl_zeros);

figure;
bode(Alt_Pitch)
grid on
title('Crop Duster Frequency Response of Alteration 1')

% Changing a variable in num + denom (A1 ----> A4)

A4 = -3.50; A2 = 0.9354; A3 = 0.2182;
B1 = -2.2796; B2 = -0.4146; B3 = 2.8334;

s = tf('s');
Alt_Pitch2 = (((B3*s) + (B1*A3 - A4*B3)) / ((s^3) - ((B4+A4)*s^2) + ((A4*B4 - A2*B1)*s)));
STP = 1; % 1 rad (57.3 deg) step response input amount of step in radians

% Open Loop Step Response

t = [0:0.01:10]; %10 sec run time
figure;
step(STP*Alt_Pitch2,t);
axis([0 10 0 0.8]);
ylabel('Pitch Angle (rad)');
title('Crop Duster Altered 2 Open Loop Step Response');
ol_poles = pole(Alt_Pitch2);
disp('Altered 2 Open Loop Poles:'); disp(ol_poles)

% Closed Loop Step Response

cl = feedback(Alt_Pitch2,1);
figure;
step(STP*cl);
ylabel('Pitch Angle (rad)');
title('Crop Duster Altered 2 Closed Loop Step Response');
cl_poles = pole(cl);
cl_zeros = zero(cl);
disp('Altered 2 Closed Loop Poles:'); disp(cl_poles);
disp('Altered 2 Closed Loop Zeros:'); disp(cl_zeros);

figure;
bode(Alt_Pitch2)
grid on
title('Crop Duster Frequency Response of Alteration 2')

% Boeing 747 Aircraft

A1 = -0.425; A2 = 0.99; A3 = -0.267;
B1 = -3.45; B2 = -0.875; B3 = 3.33;

s = tf('s');
Pitch = (((B3*s) + (B1*A3 - A1*B3)) / ((s^3) - ((B2+A1)*s^2) + ((A1*B2 - A2*B1)*s)));
STP = 1; % 1 rad (57.3 deg) step response ; input amount of step in radians

% Open Loop Step Response

t = [0:0.01:10]; %10 sec run time
figure;
step(STP*Pitch,t);
axis([0 10 0 0.8]);
ylabel('Pitch Angle (rad)');
title('747 Open Loop Step Response');
ol_poles = pole(Pitch);
disp('Open Loop Poles:'); disp(ol_poles)

% Closed Loop Step Response

cl = feedback(Pitch,1);
figure;
step(STP*cl);
ylabel('Pitch Angle (rad)');
title('747 Closed Loop Step Response');
cl_poles = pole(cl);
cl_zeros = zero(cl);
disp('Closed Loop Poles:'); disp(cl_poles);
disp('Closed Loop Zeros:'); disp(cl_zeros);

k = dcgain(cl);

figure;
bode(Pitch)
grid on
title('747 Frequency Response of Pitch Control')

Routh-Hurwitz Stability and PID Root Locus

```
% Criteria based on Routh-Hurwitz stability criterion
```

```
P = 1.3; % Proportional Gain < 1.3113 : 1.3
I = 0.4; % Integral Gain > 0 : 0.4
D = 1.2; % Derivative Gain > 0.767 : 1.2
```

```
LHS = (2.8334*I + 4.4865*P) * ((2.8334*D - 2.1736) * (4.4865*D + 2.8334*P + 2.8616) - (2.8334*I + 4.4865*P));
RHS = (2.8334*D - 2.1736)^2 * (4.4865*I);
```

```
equal = LHS > RHS
```

```
s = tf('s');
G = (2.8334*s + 4.4865) / (s^3 - 2.1736*s^2 + 2.8616*s);
C = pid(P, I, D);
T_cl = feedback(C*G, 1);
cl_poles = pole(T_cl);
[cl_num, cl_den] = tfdata(T_cl, 'v');
cl_roots = roots(cl_den);
```

```
disp('Closed-loop poles (from pole()):');
disp(cl_poles);
disp('Roots of the characteristic polynomial (from roots() on denominator):');
disp(cl_roots);
```

```
Settling_time = 4/abs(max(real(cl_roots)))
```

```
[~, idx_dom] = max(real(cl_poles));
dom_pole = cl_poles(idx_dom);
```

```
if imag(dom_pole) ~= 0
    sigma = -real(dom_pole);
    wd     = abs(imag(dom_pole));
    wn     = sqrt(sigma^2 + wd^2);
    zeta    = sigma / wn;

    Mp = exp(-pi*zeta / sqrt(1 - zeta^2)) * 100;

    disp(['Estimated percent overshoot: ', num2str(Mp), ' %']);
else
    disp('Dominant pole is real → no overshoot (monotonic response).');
end

DC_gain = dcgain(T_cl);

ess = (1 - DC_gain);

disp(['Steady-state error (unit step): ', num2str(ess)]);

L = C*G;

figure;
rlocus(L); hold on;
plot(real(cl_poles), imag(cl_poles), 'kd', 'MarkerSize', 8, 'Linewidth', 2);
grid on;
legend('Root locus of L(s)', 'Closed-loop poles (K = 1)', 'Location', 'northwest');
[k_poles, kv] = rlocus(L, 0:0.01:5);
idx = find(abs(kv-1) < 1e-6);
k_poles(:, idx)
```

Root Locus with the Inclusion of Sensor Dynamics

```
s = tf('s');
H = 1 / (0.5*s + 1);
G = (2.8334*s + 4.4865) / (s^3 - 2.1736*s^2 + 2.8616*s);
C = pid(P, I, D);
T_cl = feedback(C*G*H, 1);
cl_poles = pole(T_cl);
[cl_num, cl_den] = tfdata(T_cl, 'v');
cl_roots = roots(cl_den);

disp('Closed-loop poles (from pole()):');
disp(cl_poles);
disp('Roots of the characteristic polynomial (from roots() on denominator):');
disp(cl_roots);

Settling_time = 4/abs(max(real(cl_roots)))

[~, idx_dom] = max(real(cl_poles));
dom_pole = cl_poles(idx_dom);

if imag(dom_pole) ~= 0
    sigma = -real(dom_pole);
    wd     = abs(imag(dom_pole));
    wn     = sqrt(sigma^2 + wd^2);
    zeta   = sigma / wn;
```

```
Mp = exp(-pi*zeta / sqrt(1 - zeta^2)) * 100;

disp(['Estimated percent overshoot: ', num2str(Mp), ' %']);
else
    disp('Dominant pole is real → no overshoot (monotonic response).');
end

DC_gain = dcgain(T_cl);

ess = (1 - DC_gain);

disp(['Steady-state error (unit step): ', num2str(ess)]);

L = C*G*H;

figure;
rlocus(L); hold on;
plot(real(cl_poles), imag(cl_poles), 'kd', 'MarkerSize', 8, 'Linewidth', 2);
grid on;
legend('Root locus of L(s)', 'Closed-loop poles (K = 1)', 'Location', 'northwest');
[k_poles, kv] = rlocus(L, 0:0.01:5);
idx = find(abs(kv-1) < 1e-6);
k_poles(:, idx)
hold off
```

Root Locus Plots For 747

```
%% Root Locus for 747 Pitch Axis with Various Controllers
clc; clear; close all;

s = tf('s');

% 747 pitch-axis
Gp = (-3.33*s - 0.494) / (s^3 + 1.30*s^2 + 3.79*s);

% Control column gain Gc
K = 32; % from K vs Mach curve (lb/deg)
Gc_gain = -2/(57.3*K);

C_P = 1; % P
C_I = 1/s; % I
C_D = s; % D
C_PI = (s + 1)/s; % PI
C_PD = (s + 1); % PD
C_PID = (s^2 + s + 1)/s; % PID

zeta = 0.5912; % 10% OS
theta = acos(zeta);
slope = tan(theta);
Ts = 16;
sigma = -4/Ts; % vertical line for Ts = 8 s

function addSpecs(slope, sigma, ylimit)
    hold on;
    x = -10:0.1:0;
    plot(x, slope*x, 'm--','LineWidth',1.2); % damping line +
    plot(x, -slope*x, 'm--','LineWidth',1.2); % damping line -
    plot([sigma sigma], ylimit, 'c--','LineWidth',1.2); % Ts line
end
```

```
% P controller
L_P = Gc_gain * C_P * Gp;
figure;
rlocus(L_P); grid on; hold on;
addSpecs(slope, sigma, [-10 10]);
title('Root Locus - P Controller');

% I controller
L_I = Gc_gain * C_I * Gp;
figure;
rlocus(L_I); grid on; hold on;
addSpecs(slope, sigma, [-10 10]);
title('Root Locus - I Controller');

% D controller
L_D = Gc_gain * C_D * Gp;
figure;
```

```
rlocus(L_D); grid on; hold on;
addSpecs(slope, sigma, [-10 10]);
title('Root Locus - D Controller');

% PI controller
L_PI = Gc_gain * C_PI * Gp;
figure;
rlocus(L_PI); grid on; hold on;
addSpecs(slope, sigma, [-10 10]);
title('Root Locus - PI Controller');

% PD controller
L_PD = Gc_gain * C_PD * Gp;
figure;
rlocus(L_PD); grid on; hold on;
addSpecs(slope, sigma, [-10 10]);
title('Root Locus - PD Controller');

% PID controller
L_PID = Gc_gain * C_PID * Gp;
figure;
rlocus(L_PID); grid on; hold on;
addSpecs(slope, sigma, [-10 10]);
title('Root Locus - PID Controller');
```

```

clc; clear; close all;
s = tf('s');

% 747 pitch-axis
Gp = (-3.33*s - 0.494) / (s^3 + 1.30*s^2 + 3.79*s);

% Control-column gain
K = 32; % from K-Mach curve
Gc_gain = -2/(57.3*K);

G = Gc_gain * Gp;

zeta_des = 0.55; % damping ratio ~ 10-15% OS
Ts_des = 16; % seconds
wn_des = 4/(zeta_des*Ts_des);
wc_des = wn_des;

% Design a PID controller Ts ~ 16 s
C_pid = pidtune(G, 'pid', wc_des);

Kp = C_pid.Kp;
Ki = C_pid.Ki;
Kd = C_pid.Kd;

C_pid

C_pid =

      1
Kp + Ki * --- + Kd * s
      s

with Kp = 437, Ki = 397, Kd = 64.5

Continuous-time PID controller in parallel form.
Model Properties

T_cl = feedback(C_pid*G, 1);
figure;
step(T_cl);
grid on;
title('Closed-loop step response with designed PID');

```

Calculating 747 PID Values

References

Control Tutorials for MATLAB and Simulink. (n.d.). *Aircraft pitch: System modeling*. University of Michigan.

Retrieved November 19, 2025, from

<https://ctms.engin.umich.edu/CTMS/index.php?example=AircraftPitch§ion=SystemModeling>

OpenAI. *ChatGPT* (2025). <https://chat.openai.com/>

R. K. Heffley and W. F. Jewell, "Aircraft Handling Qualities Data," NASA CR-2144, National Aeronautics and Space Administration, 1972. <https://ntrs.nasa.gov/api/citations/19730003312/downloads/19730003312.pdf>

Franklin, Gene F, et al. *Feedback Control of Dynamic Systems*. Boston Pearson, 2020.